4 Environment

UNIT SPECIFICS

Through this unit we have discussed the following aspects:

- ***Environment, Global Warming and Climate Change***
    - *Global warming phenomena and GHG emissions,*
    - *Pollution Mitigation – Measures and Approaches,*
    - *Non-stationarity*
- ***Environmental Metrics and Monitoring***
    - *Environmental Monitoring*
    - *Global Climate Indicators and Essential Climate Variable (ECV)*
    - *Environmental Performance Index Indicators*
    - *Other Sustainability measures*
- ***Innovations and methodologies for ensuring Sustainability***
    - *Environmental Impact Assessment (EIA)*
    - *Life cycle Assessment (LCA)*
    - *Strategic Environmental Assessment (SEA)*

Besides giving a large number of multiple choice questions as well as questions of short and long answer types marked in two categories following lower and higher order of Bloom's taxonomy, a list of references and suggested readings are given in the unit so that one can go through them for practice.

There is a "Know More" section, which has been carefully designed so that the supplementary information provided in this part becomes beneficial for the users of the book. It is important to note that for getting more information on various topics of interest some QR codes have been provided which can be scanned for relevant supportive knowledge This section mainly highlights applications of the subject matter for our day-to-day real life or/and industrial applications on variety *of aspects, case study related to environmental, sustainability, social and ethical issues whichever applicable, and finally inquisitiveness and curiosity topics of the unit.*

RATIONALE

This foundational unit on environment and the impact associated with it, offers understanding of the various views on the state of the environment at present; and provides knowledge on mitigation strategies and policies, metrices and indicators, and most importantly, methodologies and tools that are applied to assess environmental impact, to empower the civil engineer in practice.

UNIT OUTCOMES

List of outcomes of this unit is as follows:

U4-O1: *Understanding of Global warming phenomena, Climate change and Pollution Mitigation*

U4-O2: *Knowledge of Environmental Monitoring & Metrics*

U4-O3: *Knowledge on innovations and methodologies for ensuring Environmental Sustainability*

Unit-4 Outcomes	***EXPECTED MAPPING WITH COURSE OUTCOMES*** *(1- Weak Correlation; 2- Medium correlation; 3- Strong Correlation)*						
	CO-1	***CO-2***	***CO-3***	***CO-4***	***CO-5***	***CO-6***	***CO-7***
U4-O1	*2*	*2*	*3*	*1*	*2*	*3*	*2*
U4-O2	*3*	*3*	*3*	*1*	*3*	*3*	*2*
U4-O3	*3*	*3*	*3*	*1*	*3*	*3*	*2*

"A clean, healthy and sustainable environment is a human right" – was adopted as a landmark resolution by The United Nations Human Rights Council in October 2021. India's Environment (Protection) Act, 1986, Section 2, defines Environment as follows; "*environment includes water, air and land and the inter-relationship which exists among and between water, air and land, and human beings, other living creatures, plants, micro-organism and property*". This Act was instituted by the Govt. of India following the nation's participation in the UN Conference on the Human Environment, Stockholm in June 1972, to prioritise the protection and improvement of environment, and prevent hazards to human beings, other living creatures, plants and property. Climate change and Global warming are commonly used today, almost interchangeably, to imply hazardous impacts on all life and argue the need for Environmental consideration and sustainable consumption.

4.1 ENVIRONMENT, GLOBAL WARMING AND CLIMATE CHANGE

The term **'global warming'** refers to the long-term heating of the Earth's surface observed since the beginning of the Industrial Era due to 'greenhouse effect' caused by human activities, primarily attributed to burning of fossil fuels and industrial processes, and deforestation which leads to significant greenhouse gas emissions. This term should not be used interchangeably with Climate change, as the latter refers to the long-term change in the average weather patterns – temperature, precipitation, wind and tidal patterns, and is not limited to the adverse effects of human activities alone. Natural causes such as, volcanic activity, cyclical ocean patterns, orbital changes, may also contribute to climate change. **Climate change** has far-reaching consequences beyond temperature increase, as it affects ecosystems, agriculture, water availability, human health, and socio-economic systems. Key indicators of Climate change are; frequency and severity changes in extreme weather such as hurricanes, heatwaves, wildfires, droughts, floods, and precipitation; ice loss at Earth's poles and in mountain glaciers; rising sea levels; cloud and vegetation cover changes, as well as the global land and ocean temperature increases. It maybe oversimplified to state that global warming leads to climate change.

4.1.1 Global warming phenomena and GHG emissions

The phenomenon of global warming occurs when greenhouse gases or GHG, such as, carbon-dioxide (CO_2), chlorofluorocarbons, water vapour, methane (CH_4), and nitrous oxide (N_2O), which are heat-trapping pollutants create a layer in the Earth's atmosphere which disallows the solar radiation emitted by the earth's surface to escape and instead absorbs the heat, thereby increasing the surface temperature. This **greenhouse effect** is a natural process that is essential for maintaining Earth's habitable temperature. However, human activities, especially the burning of fossil fuels (coal, oil, and natural gas), deforestation, and industrial processes, have significantly increased greenhouse gas emissions over the past century. The burning of fossil fuels for electricity generation, transportation, and industrial processes is the primary source of CO_2 emissions. Agricultural practices, such as livestock production and rice cultivation, contribute to CH_4 and N_2O emissions. As the concentration of greenhouse gases increases, more heat is trapped within the Earth's atmosphere, causing a rise in average global temperatures, in turn, leading to the melting of polar ice caps and rising sea-level. Increased CO_2 absorption by the oceans can lead to ocean acidification, which have detrimental effects on marine life, including coral reefs and shell-forming organisms. The changing temperatures can intensify extreme weather events, such as, hurricanes, heatwaves, droughts, and alter precipitation, which disrupts the ecosystem and affects plant and animal species' distribution and migration patterns.

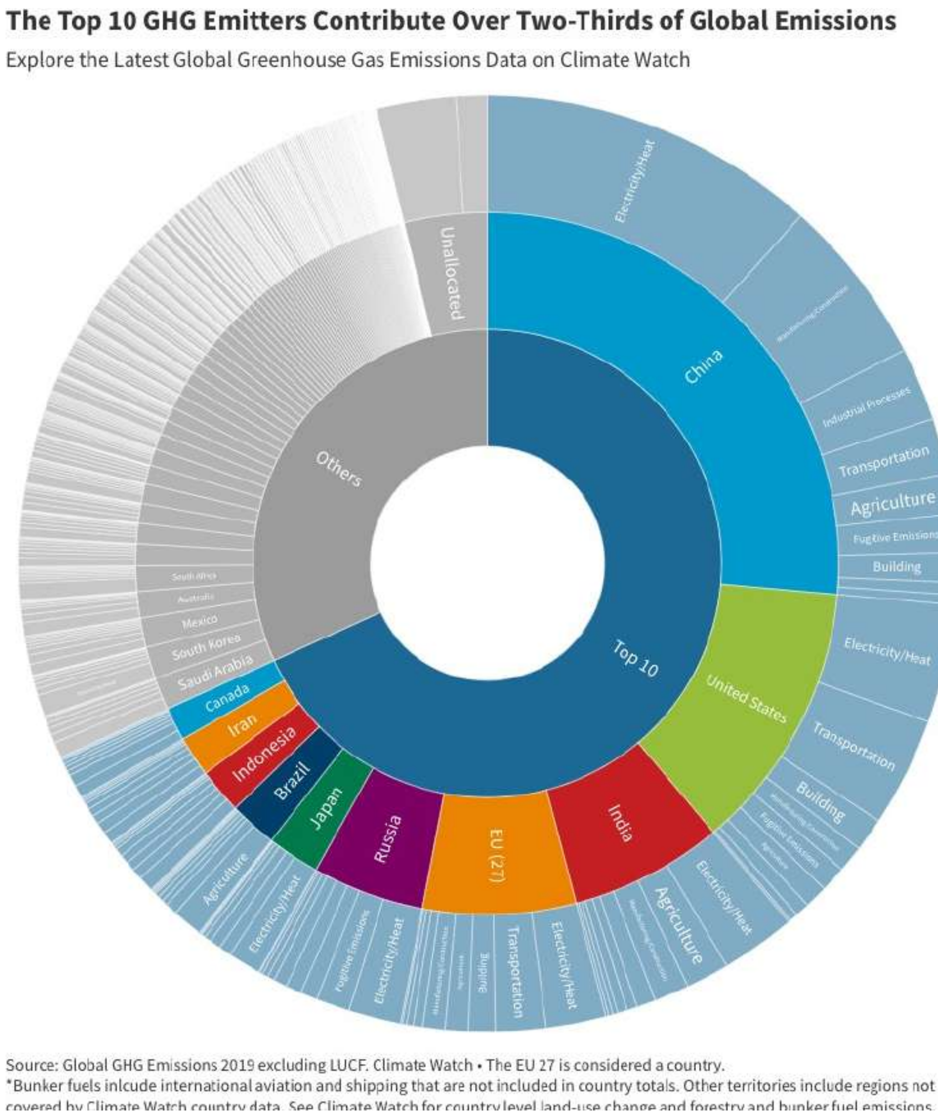

Fig. 4.1: Global GHG Emissions, 2019, by Country and Sector (source:www.climatewatchdata.org)

The biggest GHG emitting nations are China, USA, Russia, and India. The ***top 10 emitters*** (country-wise) account for over **two-thirds of annual GHG emissions**, together accounting for over 50% of the global population and 75% of the world's GDP. GHG emissions are usually measured in *CO_2 equivalent*, as CO_2 alone accounts to almost 76-78% of global GHG emissions, however the jury is still out on the strength of CO_2 to cause warming and its factual effects directly on global temperature rise, as a certain amount is essential in the atmosphere to support all life on this planet, as it is an important nutrient for trees, plants and crops. Other gases need to be converted to this unified unit by multiplying its emission to its Global Warming Potential (GWP), which is a measure of how much energy the emissions of 1 ton of a gas will absorb over a given period of time, relative to the emissions of 1 ton of carbon dioxide (CO_2). GWP acknowledges the fact that many gases are more impactful at warming Earth than CO_2, per unit mass.

The UN's Intergovernmental Panel on Climate Change (IPCC) forewarns that to limit global warming to 1.5°C, GHG emissions must peak before 2025 at the latest and decline 43% by 2030, as they indicate that crossing the 1.5°C threshold risks unleashing severe climate change impacts, including more frequent and severe droughts, heatwaves and rainfall. A landmark, multilateral climate change process – the **Paris Agreement**, binds 196 parties (nations) together to combat climate change and adapt to its effects, was adopted at the UN Climate Change Conference (COP21) in December 2015. Each Party has set 2030 targets in their Nationally Determined Contributions (NDCs) to align with the Paris Agreement temperature goal and has formulated '*long-term low greenhouse gas emission development strategies*' (LT-LEDS) that provides long-term horizon to the NDCs. Under an '*enhanced transparency framework*' (ETF), due to start in 2024, countries will be able to report transparently on actions taken and progress in climate change mitigation, adaptation measures and support provided or received, and feed the information into the **Global stocktake** which will assess the collective progress towards the long-term climate goals. Presently, this is undertaken by various organisations, such as, US Environment Protection Agency, European Environment Agency, International Energy Agency, or World Data Lab, who launched the *World Emissions Clock* at COP27.

The EPA tracks the total emissions in the United States by publishing the *Inventory of U.S. Greenhouse Gas Emissions and Sinks*, and their 2021 report notes that the primary sources of GHG emissions by economic sector are ; Transportation (28%), Power/electricity generation (25%), Industry (23%), Commercial and Residential (13%), Agriculture (10%) and Land use and Forestry (12%). Globally, Power and Energy sector is the largest contributor towards GHG emissions, and transportation and buildings have the major most activities that include both direct emissions from fossil fuel combustion, as well as indirect emissions such as use of electricity. The three sectors that stand out as the fastest-growing sources of GHG emissions since 1990 are - Industrial processes (grew by 203%); Electricity and heating, a subsector of energy (by 84%); and Transportation, also a subsector of energy (by 78%) as noted by EPA (Climate watch, 2021).

4.1.2 Pollution Mitigation

Mitigating global warming requires collective efforts and a comprehensive approach involving various sectors and stakeholders, by leveraging and building on existing mitigation measures and by incorporating sustainable approaches to ensure long-term resolution.

Emission and Pollution Mitigation Measures

Some **key strategies** for mitigating GHG emissions and pollution are as follows:

1. ***Transitioning to Renewable Energy:*** Renewable sources, such as, solar, wind, hydro, and geothermal power from fossil fuels reduces greenhouse gas emissions associated with electricity generation and decreases reliance on carbon-intensive energy sources. Another potential source for generating electricity is from nuclear energy rather than the combustion of fossil fuels.
2. ***Energy Efficiency:*** Largely, it can be achieved by; "*increasing the efficiency of existing fossil fuel-fired power plants by using advanced technologies, substituting less carbon- intensive fuels, and shifting generation from higher-emitting to lower-emitting power plants*" and by "*reducing electricity use and peak demand by increasing energy efficiency and conservation in homes, businesses, and industry*." (Contribution of Working Group III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change, chapter 6, 2021). Improving energy efficiency in transportation, buildings, and industrial processes can significantly reduce greenhouse gas emissions. This can be achieved through measures, like; switching to alternate fuels like biofuels, hydrogen, and electricity from renewable sources, or fossil fuels that are less CO_2-intensive; improving fuel efficiency and employing operating practices that minimize fuel use across all sectors. Encouraging the use of active transportation modes, like, walking and cycling, alongside improving public transportation systems and promoting it, reducing travel demand, especially office commute, through policy and urban planning to address long daily commutes will help reduce emissions from the transportation sector. In the building and allied services sectors, measure such as, improving insulation and building envelope design; incorporating efficiency strategies in heating, cooling, ventilation, and refrigeration systems; using LED lighting and passive heating and lighting to take advantage of natural sunlight; using energy-efficient appliances and electronics; incorporating energy efficiency practices into water and wastewater plants, solid waste management, and urban processes, etc. have great potential to curb emissions.
3. ***Forest Conservation and Reforestation*:** Forests act as carbon sinks, absorbing CO_2 from the atmosphere, therefore, protecting existing forests, preventing deforestation, and implementing reforestation and afforestation projects can help sequester carbon and mitigate global warming. Carbon Capture and Sequestration (CCS) where CO_2 is captured as a by-product of fossil fuel combustion, before it enters the atmosphere, and transporting to inject it deep underground at carefully selected and suitable subsurface geologic formation to be securely stored is also a potent mitigation strategy.
4. ***Sustainable Agriculture and Land Use*:** Implementing sustainable agricultural practices that reduce emissions from farming activities by emulating natural processes, promoting

agroforestry, and minimizing land-use change and deforestation in favour of carbon-rich ecosystems can contribute to global warming mitigation. "*It uses up to 56 per cent less energy per unit of crops produced, creates 64 per cent fewer greenhouse gas emissions per hectare and supports greater levels of biodiversity than conventional farming*." (UNEP, Trade and Environment Briefings: Sustainable Agriculture, 2012).

Circular Economy Approach to mitigate

The EPA endorses the "**Circular economy for all**" approach (Circular Economy systems diagram, 2019, Ellen MacArthur Foundation) to reduce waste and hazardous materials, and reuse critical minerals in manufacturing, as natural resource extraction and processing contributes to about half of all global GHG emissions (United Nations' International Resource Panel). The UN Conference on Trade and Development (UNCTAD) elaborates that circular economy is an economy where, "*all forms of waste, such as clothes, scrap metal and obsolete electronics, are returned to the economy or used more efficiently*" and "*entails markets that give incentives to reusing products, rather than scrapping them and then extracting new resources*".

Fig. 4.2: Circular Economy, UNCTAD

The **policies and strategies** developed in line with this approach for mitigating global warming are :

1. ***Waste management and Industrial Best Practices:*** Emphasizing the principles of 4R synonymous to the concept of circular economy, i.e., reduce, reuse, recycle and remanufacture, can help reduce emissions associated with, raw material extraction through conservation, across the entire life of the product from production till disposal. Encouraging industries to adopt cleaner production techniques, such as using less toxic materials, implementing energy-efficient processes, and optimizing resource usage, can help reduce pollution emissions and minimize environmental impacts.
2. ***Sustainable consumption and production:*** Individuals can contribute to global warming mitigation by making sustainable choices in their daily lives, which includes conserving energy, reducing food waste, supporting eco-friendly products, and advocating for sustainable practices. SDG 12 is about ensuring sustainable consumption and production patterns, as present unsustainable behaviour is the root cause of pollution, biodiversity loss and climate change. This is also key to sustain the livelihoods of current and future generations. The UN further adds that "*Governments and all citizens should work together to improve resource efficiency, reduce waste and pollution, and shape a new circular economy*." Moreover, raising awareness about pollution-related issues, educating the public on the importance of pollution prevention, and promoting environmentally friendly behaviours can lead to positive changes in individual and collective actions. "*Around 90 per cent of countries report that education for sustainable development and global citizenship education are at least partially mainstreamed in national education laws and policies, curricula, teacher education or student assessments in primary and secondary school.*" (SDG Report 2022). The UNCTAD is also working on research into 'incorporating sustainability into consumer protection policy aims to make a contribution to the UNCTAD RPP Research and Policy Analysis Area: The role of competition law and policy and consumer protection in economic and social development and poverty reduction, and more specifically the Sub-Element: The role of consumer protection in social development and poverty reduction.' The UN also organizes the Climate Action Summit, a high-level event that brings together global leaders, businesses, and civil society to enhance climate ambition and accelerate action, and has UN Environment's vision for Environmental Education & Training for Sustainable Development (EETSD).
3. ***International Cooperation, Environmental Rights and Rule of Law:*** Governments play a crucial role in implementing and enforcing environmental regulations and policies that set pollution control standards and promote sustainable practices, which may include emission limits, waste disposal guidelines, pollution prevention requirements, and ensuring environmental justice. International agreements, such as the Paris Agreement for climate change mitigation, facilitate collaboration among nations to address pollution issues on a larger scale, supported by bodies of UN such as the UN Environment Program (UNEP).
   The core objectives of the UNEP are to serve as an authoritative advocate for the global environment, to support governments in setting the global environmental agenda, and to promote the coherent implementation of the environmental dimension of sustainable development within the UN system, at the global and regional levels. One such means is to promote and support development and implementation of international environmental law. In 2013, UNEP's Governing Body adopted Decision 27/9, on *Advancing Justice, Governance*

and Law for Environmental Sustainability, to establish the term '**Environmental rule of law**'. The Environmental rule of law is defined as "*the rule of law as a principle of governance in which all persons, institutions and entities, public and private, including the State itself, are accountable to laws that are publicly promulgated, equally enforced and independently adjudicated, and which are consistent with international human rights norms and standards*." (Report on the Rule of Law and Transitional Justice in Conflict and Post-Conflict Societies). International environmental law is "*a branch of public international law - a body of law created by States for States to govern problems that arise between States. It is concerned with the attempt to control pollution and the depletion of natural resources within a framework of sustainable development*." (UNEP, 2017).
The UNEP also hosts the Climate Technology Centre and Network (CTCN), the implementation arm of the Technology Mechanism of the United Nations Framework Convention on Climate Change. The Centre promotes "*the accelerated transfer of environmentally sound technologies for low carbon and climate resilient development at the request of developing countries. The CTC provides technology solutions, capacity building and advice on policy, legal and regulatory frameworks tailored to the needs of individual countries by harnessing the expertise of a global network of technology companies and institutions*". Addressing global warming requires international cooperation and the implementation of effective policies at the national and international levels. This includes setting ambitious emission reduction targets, implementing carbon pricing mechanisms, and fostering collaborations to support technology transfer and capacity building in developing countries.

4.1.3 Non-stationarity

Statistical analysis of the difference between mean CO_2 concentration (ppm) in the 1980s and mean CO_2 concentration in the 2000s, captured by Mauna Loa, Hawaii, which has the longest continuous record of direct atmospheric CO_2 measurements, showed that there has been a significant change in mean maximum temperature from decade to decade, mean maximum temperature from decade to decade for spring, mean maximum temperature from season to season, and further established that there is statistical evidence to claim that climate variables are changing over time (causeweb.org). The climate system is constantly changing across time, and these changes are usually associated with the temporal (time related) variation of the statistical properties of climatic variables, such as temperature, precipitation, wind speed, relative humidity, volcanic activity, seasonal change, and solar radiation levels. Thus, statistically it poses non-stationarity, i.e., the status of a time series whose statistical properties are changing through time, and is opposite of a stationary time series which has statistical properties or moments (e.g., mean and variance) that do not vary in time. Beuno de Mesquita, et al. (2020) define **non-stationarity** as "*a change in the relationship, either in direction or magnitude (from a significant relationship to no relationship and vice versa) between variables over time*". They further add that "*such non-stationarity is often combined with the additional hurdle of needing multivariate statistics*" as the changes in climate have become exceedingly erratic such that previous thought to be coupled climatic characteristics, such as, temperature and drought, are no longer behaving in tandem but now need to be considered as potentially independent.

4.2 ENVIRONMENTAL MONITORING & METRICS

Environmental impact is succinctly described as "*changes in the natural or built environment, resulting directly from an activity, that can have adverse effects on the air, land, water, fish, and wildlife or the inhabitants of the ecosystem. Pollution, contamination, or destruction that occurs as a consequence of an action, that can have short-term or long-term ramifications is considered an environmental impact.*" (Abdallah, 2017). These impacts require a system or standard of measurement, i.e., metric, that indicates the state of a system and measures the its behaviour. Therefore, the term '**indicator'** is commonly used to refer to these metrics for ascertaining the content and performance of a system, corresponding to any of the three dimensions of sustainability - ecological metrics, economic metrics, and sociological metrics (Sikdar,2003). However, holistic understanding, measuring and monitoring can only be achieved at the intersection of these three dimensions, as most systems are complex and are inadequately represented. Therefore, '**true sustainability metrics'** are 3-Dimensional and are classified as follows;

Group 1 (1-D): economic, ecological, and sociological indicators
Group 2 (2-D): socio-economic, eco-efficiency, and socio-ecological indicators
Group 3 (3-D): sustainability indicators

4.2.1 Environmental Monitoring

A large number of sustainability indictors have been thematically classified with respect to the 17 SDGs and discussed in earlier Unit, and there are various means for each to be measured and monitored. *The systematic collection, analysis, and interpretation of data on various aspects of the environment, involving measuring, observing, and assessing environmental indicators to understand the current state, changes, and trends in natural systems and human-induced activities* is referred to as **Environmental monitoring**. The data collected through environmental monitoring helps in assessing the effectiveness of environmental policies, identifying potential risks and hazards, and informing decision-making processes related to resource management, conservation, and pollution control. It encompasses a wide range of parameters, including air quality, water quality, soil health, biodiversity, climate variables, noise levels, and pollution levels, and involves the use of various monitoring techniques, such as sampling, remote sensing, sensor networks, and data analysis tools, to gather accurate and reliable information for environmental assessment and management.

Below enlisted are some common types of **environmental monitoring and the tools/methods** used for each:

1. ***Air Quality Monitoring:***
   - Tools: Air quality monitors (with CO_2 and O_2 sensors), air sampling equipment, Temperature and Humidity monitors, particulate matter (PM) samplers, gas analysers.

- Methods: Continuous monitoring stations, passive samplers, mobile monitoring, remote sensing, and modelling techniques.

2. ***Water Quality Monitoring:***
   - Tools: Water quality sensors, water samplers, pH meters, turbidity meters, dissolved oxygen meters, Conductivity probes.
   - Methods: Grab sampling, automated water quality monitoring stations, remote sensing, and satellite imagery.
3. ***Soil Monitoring:***
   - Tools: Soil sampling equipment, soil moisture sensors, pH and nutrient analysers.
   - Methods: Soil sampling and analysis, soil moisture monitoring, soil fertility testing, spectrometry (to measure contamination in soils), remote sensing (to monitor salinity in soils).
4. ***Biodiversity Monitoring:***
   - Tools: Camera traps, acoustic sensors, GPS devices, binoculars, species identification guides.
   - Methods: Field surveys, transect sampling, mark and recapture techniques, bio-acoustic monitoring, remote sensing.
5. ***Noise Monitoring:***
   - Tools: Sound level meters, noise dosimeters, acoustic monitoring systems.
   - Methods: Point measurements, continuous monitoring stations, noise mapping, community noise surveys.
6. ***Pollution & Waste Monitoring:***
   - Tools: Gas analysers, spectrometers, emission measurement devices, pollutant-specific sensors.
   - Methods: Source emission sampling, ambient air monitoring, industrial effluent monitoring, pollution source tracking, production and consumption patterns.
7. ***Radiation Monitoring:***
   - Tools: Radiation detectors, dosimeters, Geiger-Muller counters.
   - Methods: Radiation monitoring stations, personal monitoring, environmental sampling, remote sensing.
8. ***Climate Monitoring:***
   - Tools: Weather stations, temperature loggers, rainfall gauges, anemometers, radiometers.
   - Methods: Meteorological observations, climate stations, satellite-based measurements, climate modelling.

4.2.2 Global Climate Indicators and Essential Climate Variable (ECV)

The *World Meteorological Organization (WMO)* **State of the Global Climate** uses seven Global Climate Indicators - Surface Temperature and Ocean Heat (Temperature and Energy); Atmospheric CO2 (Atmospheric composition); Ocean Acidification and Sea-level (Ocean and Water); Glaciers and Arctic and Antarctic Sea Ice Extent (Cryosphere) —to monitor the domains most relevant to climate change, "*without reducing climate change to only temperature*". In addition to the seven headline indicators, a supplementary set of subsidiary indicators is complimentarily available to offer information and contribute to a more comprehensive and detailed depiction of the evolving trends in their respective areas. These indicators are derived from analysis and interpretation of fundamental measurements or parameters for characterizing and understanding the Earth's climate system.

An **Essential Climate Variables (ECV)** is defined by WMO as "*a physical, chemical or biological variable or a group of linked variables that critically contributes to the characterization of Earth' s climate*". These are compiled into datasets to, provide empirical evidence that helps understand and predict the evolution of climate, and assess risks and enable attribution of climate events to underlying causes. EVC datasets *(refer Fig. 3)* further guide mitigation and adaptation measures, and supports the United Nations Framework Convention on Climate Change (UNFCCC) and Intergovernmental Panel on Climate Change (IPCC) efforts for assessment and creating climate services. Assessments provide policymakers and the public with synthesized information on the state of the climate system, projected future changes, and associated impacts, by quantifying the observed changes and attributing them to human activities. Climate services provide tailored climate information and predictions to various sectors, such as, agriculture, water resources, energy, and disaster management, and helps develop climate models, downscaling techniques, and decision support tools.

Fig. 4.3 : Essential Climate Variables, WMO (source : https://gcos.wmo.int/en/essential-climate-variables)

The National Centers for Environmental Information (NCEI) of the U.S. National Oceanic and Atmospheric Administration (NOAA) and the U.S. GCOS Program at NCEI, maintains the Global Observing Systems Information Center (GOSIC) which provides further background, definitions, requirements, network information and data sources for ECVs. These variables are monitored as per the GCOS Climate Monitoring Principles (*agreed by the Committee on Earth Observation Satellites (CEOS) and adopted by the Congress of the World Meteorological Organization (WMO) and Conference of the Parties (COP-9, 2003) to the UNFCCC*), which provide a standardized framework for monitoring and measuring key climate parameters consistently across different regions and over extended periods. The ECVs also help in validating and calibrating climate models, ensuring they accurately represent the observed climate variability and trends.

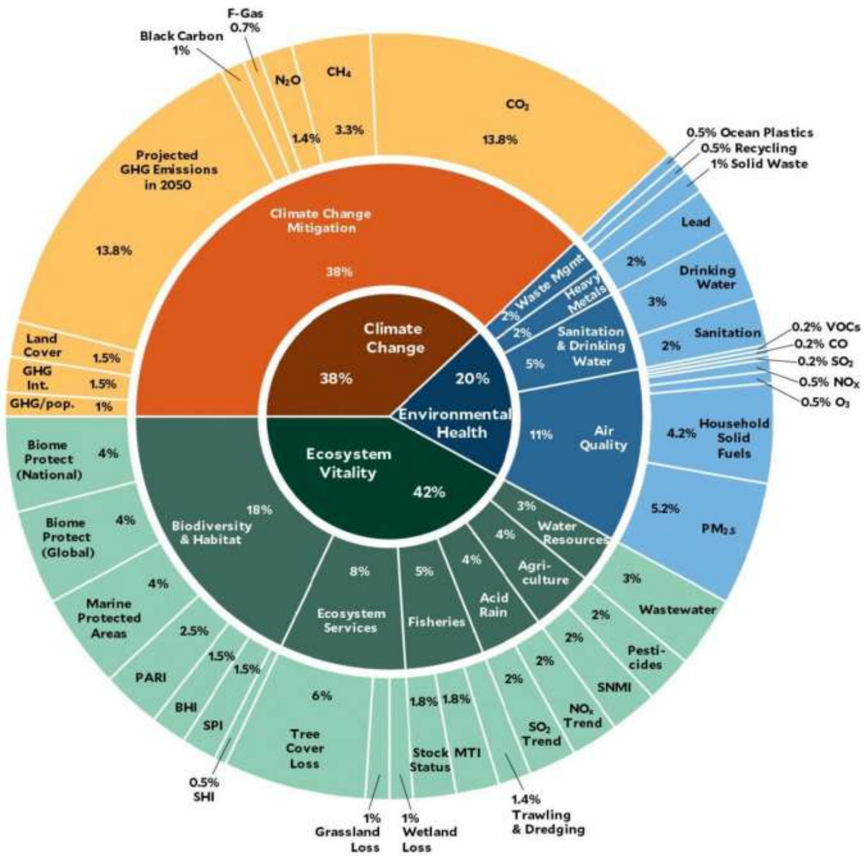

Fig. 4.4 : 2022 EPI Framework (source : https://epi.yale.edu)

4.2.3 Environmental Performance Index

The **Environmental Performance Index (EPI)** is a comprehensive assessment that utilizes data-driven analysis to present an overview of sustainability worldwide, by employing 40 performance indicators across 11 issue categories, grouped under three key themes – Climate Change, Environmental Health and Ecosystem Vitality; and ranks 180 countries based on their performance in these three themes. The **40 indicators** serve as a national-scale measure of a country's progress towards established environmental policy targets, and offer a valuable means to identify challenges, establish goals, monitor trends, comprehend outcomes, and recognize effective policy practices. By utilizing accurate data and evidence-based analysis, government officials can refine their policy agendas, engage in meaningful dialogues with stakeholders, and maximize the impact of environmental investments. The EPI contributes to the pursuit of the SDGs as a policy tool and showcases successful initiatives and practices that countries can adopt from their top-performing counterparts.

The EPI ***scores as a whole*** to help to highlight countries excelling in sustainability and draw attention to those lagging, while the detailed disaggregated data that accompanies it provides a more nuanced tool for pinpointing policy deficiencies and anomalies. It is correlated with country wealth, although some countries outperform their economic peers while others lag. However, there are several criticisms to this Index as it relies on a limited set of indicators and metrics to assess a country's environmental performance. This tends to oversimplify complex environmental issues and fails to capture the full scope of a country's sustainability efforts, particularly at the interplay between social, economic, and environmental dimensions of sustainability.

4.2.4 Other Sustainability Measures

There are other foci to sustainability measurement beyond environment and some of these are;

Social Impact Assessment: Evaluates the social and cultural effects of a project or initiative, including factors like employment, community well-being, human rights, and stakeholder engagement.

Economic Indicators: Assess the economic sustainability of a system or project, including metrics such as cost-benefit analysis, return on investment (ROI), and economic value added (EVA).

Corporate Social Responsibility (CSR) Reporting: Measures and reports on an organization's social, environmental, and economic performance, often using globally recognized frameworks like the Global Reporting Initiative (GRI) standards.

Sustainable Procurement: Evaluates the sustainability performance of suppliers and ensures that products and services are procured from environmentally and socially responsible sources.

4.3 INNOVATIONS AND METHODOLOGIES FOR SUSTAINABILITY

Sound methodologies, and decision-making or implementation tools, such as, EIA , SEA, LCA, are of great importance for ensuring overall environmental sustainability.

EIA (Environmental Impact Assessment), Life cycle Assessment (LCA) and SEA (Strategic Environmental Assessment) are **systematic approaches and tools** used to gather and evaluate environmental information before making decisions in the development process. These approaches involve making predictions about how the environment will be affected by various alternative actions and providing guidance on managing those changes if a particular alternative is chosen and implemented. EIA primarily focuses on evaluating the potential environmental impacts of specific physical developments like highways, power stations, water resource projects, and large-scale industrial facilities. LCA provides a framework for measuring the environmental impact of a product across its life and has several tools such as, GaBi, openLCA, SimaPro and Ecochain Mobius. On the other hand, SEA concentrates on assessing proposed actions at a broader level, such as new or modified laws, policies, programs, and plans. It is common for physical developments and projects to be outcomes of the implementation of policies or plans. Thus, an *integrated approach is recommended* (UNEP).

4.3.1 Environmental Impact Assessment (EIA)

The goal of an EIA is to identify and assess the potential positive and negative impacts on the environment, as well as the social and economic aspects associated with the project. The EIA Methodology is as follows:

1. **Screening**: The project plan is screened for scale of investment, location and type of development and if the project needs statutory clearance.
2. **Scoping**: Then the EIA process begins with scoping, where the project's objectives, potential impacts, zone of impacts or boundaries, mitigation possibilities and need for monitoring are identified. This step involves consultations with stakeholders, including government agencies, local communities, and experts, to determine the key issues that should be considered during the assessment.
3. **Baseline Study and Data collection**: A thorough analysis of the existing environmental conditions and socio-economic aspects is conducted, to collect data on air and water quality, biodiversity, land use, cultural heritage, and other relevant parameters, at baseline. The baseline study establishes a benchmark against which potential impacts will be compared.
4. **Impact Prediction (Identification and Assessment)**: Potential environmental impacts associated with the project are identified based on the project's characteristics, such as location, size, and technology used. This step involves analysing the project's activities, processes, and interactions with the environment to determine the potential effects. The identified impacts - Positive and negative, reversible and irreversible, and temporary and permanent impacts - are assessed in terms of their magnitude, duration, and significance. This includes evaluating both the direct and indirect impacts on various environmental

components, such as air, water, soil, biodiversity, and human health. Mitigation measures and alternatives may also be considered at this stage.

5. **Mitigation measures and EIA Report:** Based on the impact assessment, potential mitigation measures are identified to avoid, minimize, or compensate for adverse environmental effects. Alternative options, including project design modifications or alternative locations, are also explored to mitigate potential impacts. An Environmental Management Plan (EMP) is developed and reported to guide the implementation, monitoring, and management of the project, which outlines the specific measures and actions to be taken to minimize or mitigate environmental impacts, or the level of compensation for probable environmental damage or loss in adverse situations, throughout the project's lifecycle.

6. **Public Hearing and Consultation**: Throughout the EIA process, public participation and consultation are crucial. Stakeholders, including local communities, NGOs, and interested parties, are given the opportunity to provide input, express concerns, and raise questions related to the project and its potential impacts.

7. **Decision-Making and Approval**: The final EIA report, which includes the findings of the assessment, proposed mitigation measures, and alternatives, is submitted to the relevant regulatory authorities, such as, the Impact Assessment Authority along with the experts, consultants and the project-in-charge, for decision-making. The decision may involve approving the project with conditions, rejecting it, or requesting further information or modifications.

8. **Monitoring and Implementation of EMP**: Once the project is approved and implemented, ongoing monitoring and evaluation are conducted to ensure compliance with the proposed mitigation measures and environmental commitments outlined in the EMP. This step helps to assess the effectiveness of the EIA process and enables adaptive management to address any unforeseen impacts.

9. **Assessment of Alternatives, Delineation of Mitigation Measures and EIA Report**: Each project should consider and compare various potential alternatives in terms of their environmental characteristics, which should encompass both the location of the project and the technologies employed in its processes. After a thorough review of the alternatives, a mitigation plan should be developed for the chosen option. This plan should be accompanied by an Environmental Management Plan (EMP) that provides guidance to the project proponent on implementing environmental enhancements.

10. **Risk assessment**: The EIA process also includes conducting an inventory analysis and assessing the hazard probability and index as part of EIA procedures.

The EIA process may vary in detail and requirements depending on the country, jurisdiction, and specific project characteristics, however, the core objective remains the same.

4.3.2 Life Cycle Assessment (LCA)

LCA is a methodology used to evaluate the environmental impacts associated with a product, process, or system throughout its entire life cycle. There are two different approaches to conducting LCA - the '***SETAC/EPA Framework for Life Cycle Assessment'***, jointly developed by the U.S. Environmental Protection Agency (EPA) and the Society of Environmental Toxicology and Chemistry (SETAC); and **ISO 14040**, which is an international standard developed by the International Organization for Standardization (ISO). The choice between the two may depend on factors such as regional applicability, regulatory requirements, and the specific needs of the LCA study.

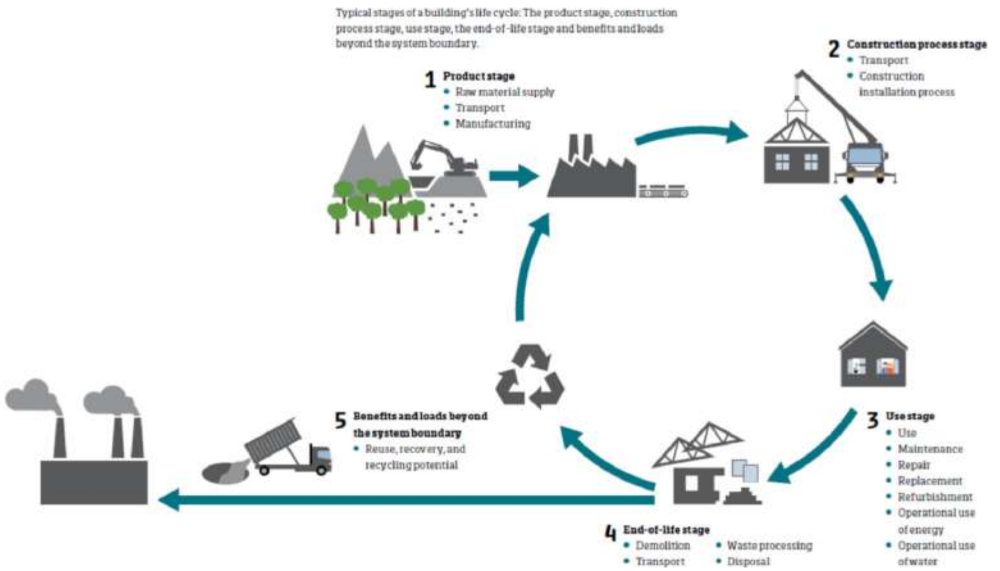

Fig. 4.5 : Lifecycle of a Building
(source : The Use of LCA for Environmental Building Assessment : A Vision of the Future – White Paper, 2017, EURIMA)

While they share common principles and goals, there are few differences, such as;

- ***Coverage and Scope:***
    - The SETAC/EPA framework emphasizes the need to consider the full life cycle of a product or system, from cradle to grave, including raw material extraction, manufacturing, use, and end-of-life.
    - ISO 14040 also advocates for a comprehensive life cycle perspective but allows flexibility in defining the scope of the study based on the intended application and goals.

- ***Methodology:***
  - The SETAC/EPA framework provides more specific guidance on conducting impact assessment, including midpoint and endpoint approaches.
  - ISO 14040 provides general principles and guidelines but does not prescribe specific impact assessment methodologies, allowing flexibility in selecting appropriate methods.
- ***Application:***
  - The SETAC/EPA framework is commonly used in North America and may be more aligned with regulatory requirements in that region.
  - ISO 14040 is an international standard and is widely adopted globally, accommodating different regional contexts and requirements.

The ISO 14040 outlines a **four-step process** for conducting an LCA, as elaborated below;

1. **Goal and Scope Definition**
   - *Define the purpose and boundaries of the LCA study*, including the product or system to be assessed and the specific environmental impacts of interest.
   - *Identify the functional unit*, which quantifies the performance of the product or system being evaluated.
   - *Determine the system boundaries*, considering all relevant life cycle stages (from raw material extraction to end-of-life disposal).
2. **Life Cycle Inventory (LCI)**
   - *Compile a comprehensive inventory* of all inputs (energy, materials, water, etc.) and outputs (emissions, waste, etc.) associated with each life cycle stage.
   - *Collect data* on resource consumption, emissions, and waste generation from primary and secondary sources, such as databases, literature, and industry-specific data, and model it into input-output flows.
3. **Life Cycle Impact Assessment (LCIA)**
   - *Evaluate the potential environmental impacts* based on the inventory data collected, upon Selection of indicators and models, Classification of Life Cycle Inventory and assigning it to our defined impact categories, and then, calculate all our equivalents, for example, Global Warming Potential (CO_2-equivalent in kg).
   - *Use impact assessment methods*, such as midpoint or endpoint approaches, to quantify the environmental effects in categories like climate change, resource depletion, human health, ecosystem quality, etc. and sum up in overall impact category totals.

4. **Interpretation**
   - *Analyse and interpret the results* to understand the implications and draw conclusions.
   - *Identify opportunities for improvement* and suggest strategies for reducing environmental impacts.
   - *Consider the limitations and uncertainties* of the study and communicate the findings accurately and transparently.

Additional considerations;

- ***Sensitivity analysis*** to assess the influence of different parameters or assumptions on the results.
- ***Reporting*** through a comprehensive report summarizing the study methodology, results, and conclusions, and ***Communicating the findings*** to stakeholders and facilitate informed decision-making.

4.3.3 Strategic Environmental Assessment (SEA)

SEA helps incorporate environmental considerations into strategic planning and policy-making processes, and ensures that environmental and social impacts are systematically assessed and integrated into decision-making, promoting sustainable development, through the following steps;

1. **Initiation and Scoping**
   - *Determine the need for SEA* and establish the objectives and scope of the assessment.
   - *Identify the key decision-makers and stakeholders* who should be involved in the SEA process.
   - *Define the legal and institutional framework* for conducting the SEA.
2. **Baseline Assessment**
   - *Collect and analyse information* about the existing environmental, social, and economic conditions.
   - *Identify the potential environmental effects* associated with the plan, policy, program, or strategy under consideration.
   - *Consider relevant environmental policies,* legislation, and international commitments.
3. **Setting Objectives and Developing Alternatives**
   - *Establish environmental objectives and targets* that align with sustainable development goals.

- *Generate a range of alternative options* or scenarios that could achieve the objectives.
- *Assess the potential environmental and social impacts* associated with each alternative.

4. Impact Assessment

- *Evaluate the potential effects* of each alternative on the environment, including direct and indirect impacts.
- *Analyse the cumulative effects* of multiple projects or activities when appropriate.
- *Identify the key environmental and social issues* that need to be addressed during implementation.

5. **Mitigation and Enhancement Measures**

- *Develop measures to avoid, minimize, or mitigate* adverse environmental and social impacts.
- *Explore opportunities* to enhance positive environmental and social outcomes.
- *Consider alternative approaches*, technologies, or management strategies that promote sustainable development.

6. Integration and Decision-Making

- *Integrate the findings* of the SEA into the decision-making process for the plan, policy, program, or strategy.
- *Communicate the results* of the assessment to decision-makers, stakeholders, and the public.
- *Consider the SEA recommendations* and findings alongside other relevant factors in the decision-making process.

7. Monitoring, Review, and Adaptation

- ***Establish a monitoring and review mechanism*** to track the implementation of the plan, policy, program, or strategy.
- ***Evaluate the effectiveness of the SEA*** recommendations and measures over time.
- ***Incorporate adaptive management practices*** to ensure continuous improvement and responsiveness to changing circumstances.

Overall, these methodologies and tools are critical in ensuring that environmental considerations are integrated into decision-making processes at different levels, from strategic planning to project implementation, by promoting sustainable development through stakeholder engagement.

UNIT SUMMARY

This unit on environmental (impact) discusses the nuances between the observable phenomena of global warming and its potential connection to overall climate change. Firstly, it elucidates the foreboding impact of pollution and the various strategies and policies for mitigation. Secondly, it explains the challenges and uncertainties in monitoring and predicting such changes due to the non-stationary nature of the variables. It further elaborates on environmental monitoring, metrics, indicators, etc to better define the state of the environment and its performance. Finally, the unit explicates the various methodologies and tools innovated to quantify environmental impact, in order to achieve sustainable development goals and instil urgency and understanding in stakeholders for immediate action.

EXERCISES

I. Multiple Choice Questions

Q. 4.1 Which of the following are greenhouse gases (GHG)?
- (a) all of the below
- (b) Methane (CH4)
- (c) Nitrous oxide (N2O)
- (d) Carbon di oxide (CO2)

Q. 4.2 What is the percentage of CO2 in the global GHG emissions?
- (a) more than 60%
- (b) 76-78%
- (c) less than 40%
- (d) 50%

Q. 4.3 Which of the following sectors is the fastest growing source of GHG emissions (since 1990) ?
- (a) Industrial processes
- (b) Transportation
- (c) Electricity and heating

(d) Agriculture and Land Use

Q. 4.4 What is the most important indicator with respect to climate change?
(a) Surface Temperature and Ocean Heat (Temperature and Energy)
(b) Ocean Acidification and Sea-level
(c) Glaciers and Arctic and Antarctic Sea Ice Extent (Cryosphere)
(d) all of the above

Q. 4.5 Which of the following are referred as 'true sustainability metrics'?
(a) Economic, ecological, and sociological indicators
(b) Sustainability indicators
(c) Socio-economic, eco-efficiency, and socio-ecological indicators
(d) Environmental Performance Index

Answers of Multiple Choice Questions: 4.1 (a) , 4.2 (b) , 4.3 (a), 4.4 (d), 4.5 (b)

II. Short and Long Answer Type Questions

Q. 4.6 What is Global Warming? Briefly explain the phenomena and its impact on environment.

Q. 4.7 What are some of the potential GHG emissions and pollution mitigation measures presently being employed?

Q. 4.8 Discuss the various types of environmental monitoring and tools/methods used for each.

Q. 4.9 What are the various types of 'metrics and indicators' for denoting the state of the environment? How are these helpful in predicting environmental performance and impact, elucidate with examples.

Q. 4.10 What are some of the innovative methodologies and tools for assessing environmental impact? Illustrate anyone that will be most useful to the profession of civil engineering.

KNOW MORE

About World Greenhouse Gas emission & World Emissions Clock

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About Sustainable agriculture

https://wedocs.unep.org/bitstream/handle/20.500.11822/25950/sustainable_agriculture.pdf?sequence=1&isAllowed=y

About Circular Economy

https://buildingcircularity.org

The Paris Agreement

https://unfccc.int/process-and-meetings/the-paris-agreement

International Environmental Law

https://wedocs.unep.org/bitstream/handle/20.500.11822/21491/MEA-handbook-Vietnam.pdf?sequence=1&isAllowed=y

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